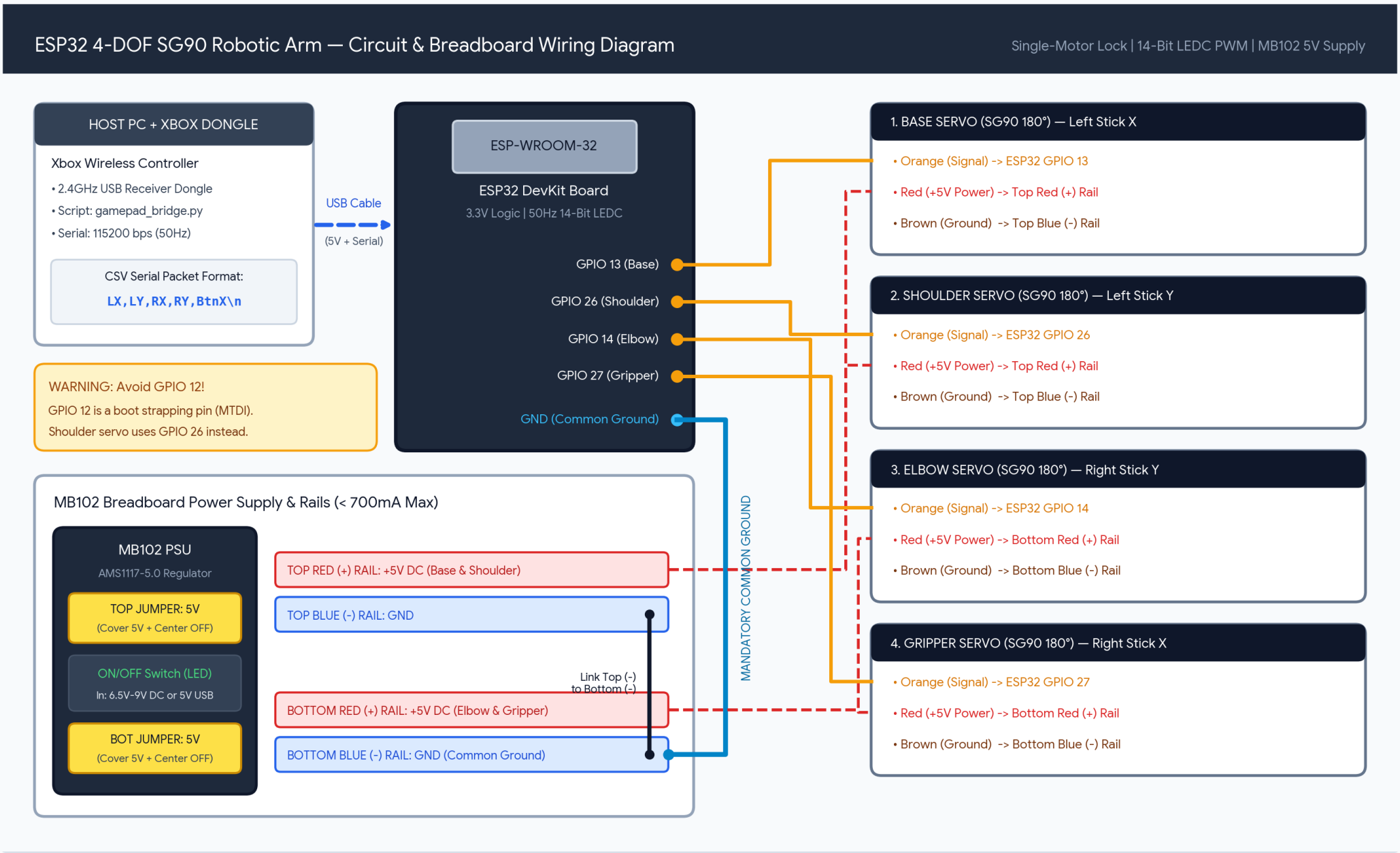


Robotic Arm Joint	Xbox Controller Input	SG90 Orange Wire (PWM Signal)	SG90 Red Wire (+5V Power)	SG90 Brown Wire (Ground)	Safe Pulse Width Range
1. Base Joint	Left Stick (Left / Right)	**ESP32 GPIO 13**	Top Red (+) Rail (5V)	Top Blue (-) Rail (GND)	750µs – 2150µs (15°–165°)

Robotic Arm Joint	Xbox Controller Input	SG90 Orange Wire (PWM Signal)	SG90 Red Wire (+5V Power)	SG90 Brown Wire (Ground)	Safe Pulse Width Range
2. Shoulder Joint	Left Stick (Up / Down)	**ESP32 GPIO 26** *(Moved from 12)*	Top Red (+) Rail (5V)	Top Blue (-) Rail (GND)	750µs – 2150µs (15°–165°)
3. Elbow Joint	Right Stick (Up / Down)	**ESP32 GPIO 14**	Bottom Red (+) Rail (5V)	Bottom Blue (-) Rail (GND)	750µs – 2150µs (15°–165°)
4. Gripper / Claw	Right Stick (Left / Right)	**ESP32 GPIO 27**	Bottom Red (+) Rail (5V)	Bottom Blue (-) Rail (GND)	750µs – 2150µs (15°–165°)

CRITICAL WIRING & POWER WARNINGS: (1) Connect an **ESP32 GND pin** to the **Breadboard Blue (-) Rail** and link the top and bottom Blue (-) rails together. (2) Set both yellow jumper caps on the **MB102** to cover the **5V** and center **OFF** pins. (3) The MB102 supplies **< 700mA max**; firmware enforces a single-motor-at-a-time movement lock and 250ms PWM auto-mute to prevent brownout resets and gear grinding.